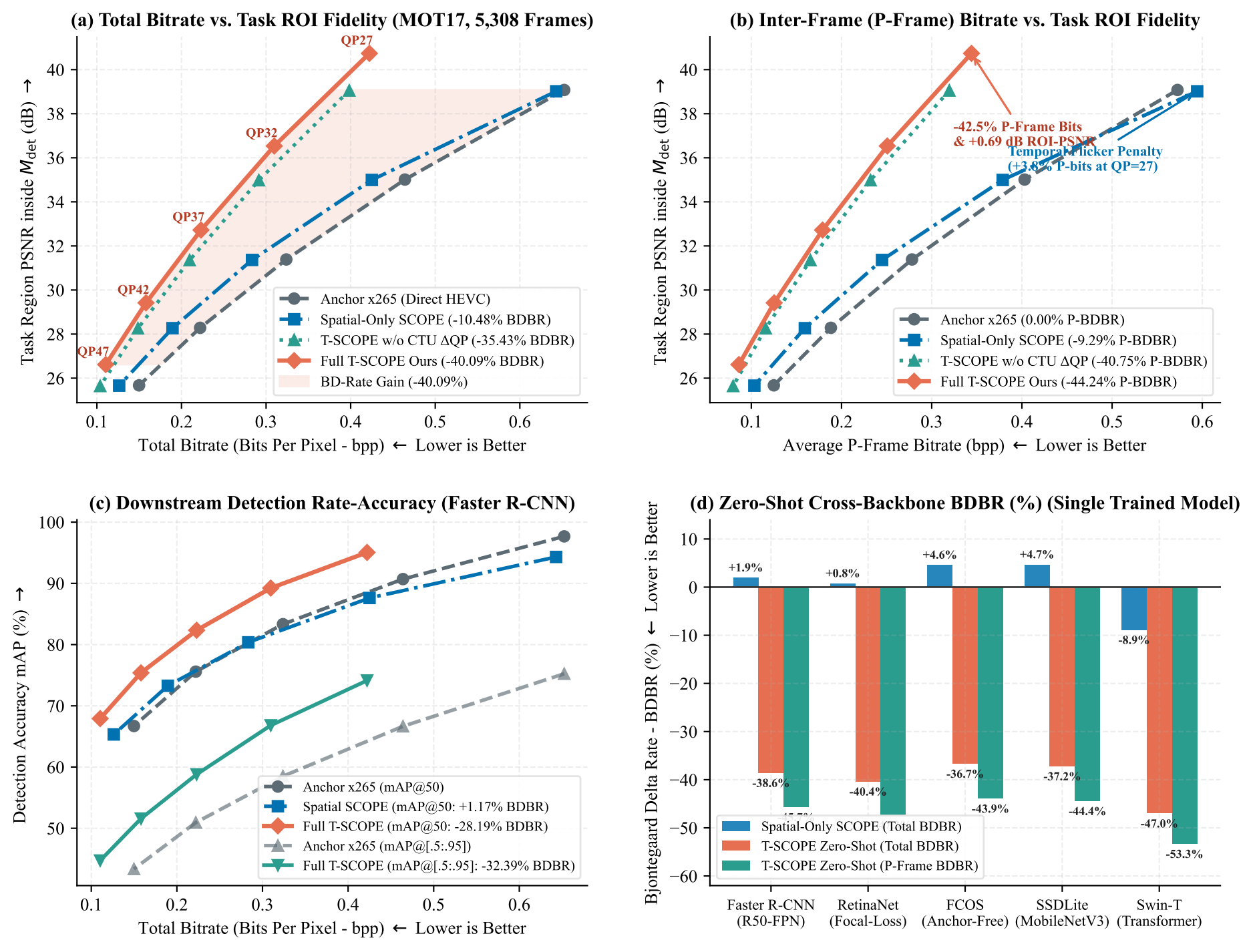

(a) Total Bitrate vs. Task ROI Fidelity (MOT17, 5,308 Frames)
Task Region PSNR inside M_{det} (dB) →
Total Bitrate (Bits Per Pixel - bpp) ← Lower is Better
QP27
QP32
QP37
QP42
QP47
Anchor x265 (Direct HEVC)
Spatial-Only SCOPE (-10.48% BDBR)
T-SCOPE w/o CTU ΔQP (-35.43% BDBR)
Full T-SCOPE Ours (-40.09% BDBR)
BD-Rate Gain (-40.09%)
(b) Inter-Frame (P-Frame) Bitrate vs. Task ROI Fidelity
Task Region PSNR inside M_{det} (dB) →
Average P-Frame Bitrate (bpp) ← Lower is Better
-42.5% P-Frame Bits
& +0.69 dB ROI-PSNR
Temporal Flicker Penalty
(+3.8% P-bits at QP=27)
Anchor x265 (0.00% P-BDBR)
Spatial-Only SCOPE (-9.29% P-BDBR)
T-SCOPE w/o CTU ΔQP (-40.75% P-BDBR)
Full T-SCOPE Ours (-44.24% P-BDBR)
(c) Downstream Detection Rate-Accuracy (Faster R-CNN)
Detection Accuracy mAP (%) →
Total Bitrate (Bits Per Pixel - bpp) ← Lower is Better
Anchor x265 (mAP@50)
Spatial SCOPE (mAP@50: +1.17% BDBR)
Full T-SCOPE (mAP@50: -28.19% BDBR)
Anchor x265 (mAP@[.5:.95])
Full T-SCOPE (mAP@[.5:.95]: -32.39% BDBR)
(d) Zero-Shot Cross-Backbone BDBR (%) (Single Trained Model)
Bjontegaard Delta Rate - BDBR (%) ← Lower is Better
+1.9%
+0.8%
+4.6%
+4.7%
-8.9%
-38.6%
-40.4%
-36.7%
-37.2%
-47.0%
-43.9%
-44.4%
-53.3%
Spatial-Only SCOPE (Total BDBR)
T-SCOPE Zero-Shot (Total BDBR)
T-SCOPE Zero-Shot (P-Frame BDBR)
Faster R-CNN (R50-FPN)
RetinaNet (Focal-Loss)
FCOS (Anchor-Free)
SSDLite (MobileNetV3)
Swin-T (Transformer)